

wear and tear mechanism and alignment problems. Everything is monitored using inertial measurement and position sensors. From these sensors we can implement a sensor-fusion algorithm. Based on this algorithm, prioritisation of call management and memory organisation helps improve the intelligence and efficiency of the robot, which can also be considered as an intelligent smart mobile robot with trolley-type material transportation system,” says Vishnu Kumar, Freelance Consultant at Embedded Systems and Mobile Robotics.

All these things determine how a robot can autonomously work in a certain environment.

As processing speed requirements are increasing day-by-day, it is critical for accessing the memory from the processor and other peripherals. At the same time, for efficient power management and consumption, certain algorithms should be improvised. Normal battery power is enough for today’s robotic needs, but in future a number of motors and multiple power handling devices will be required. Hence, thought should also be given to power management. The size and weight of the power device is also quite important.

Edge device design and manufacture

The government of India is taking a lot of initiatives to promote manufacturing under its Make in India programme. But the challenge while focusing on embedded systems or modules is that there is insufficient support for memory production and manufacturing ecosystems in India. These need to be developed.

Speakers

Dharmendra Kumar,
IoT & Digital
Business Head,
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Vishnu Kumar,
freelance consultant,
Embedded Systems
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Nikhil Bhaskaran,
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Dharmendra informs that in some other countries, research involving nanotube RAM (NRAM) is going on. So, India should also go for NRAM and adopt this technology. Technologies such as ReRAM and NRAM have been developed, but India is lagging behind with respect to these. As the development platform exists in the country, it should be utilised. “Although we are making certain efforts, we need the ecosystem. Government is taking a lot of initiatives, but they need to support the ecosystem even for design,” he says.

On a similar note, Nikhil feels that emphasis should be given more on design than on manufacture. Just as China is good at manufacturing, India is good at designing. “If we have to be a leader, then we have to mass educate on the latest technologies rather

than holding them. That is all what is required for India to become a leader in design,” he says. He thinks that to fulfil our manufacturing aspirations, we should take inspiration from the Chinese manufacturing industry while staying practical with respect to our strength, which is design.

Further insights

The next level of innovation is going to be in quantum computing. Since quantum is multi-dimensional, an intelligent chip is needed that will be able to do all the management.

On the possibility of increasing the RAM capacity in a given device, the emergence of ReRAM can help increase it, especially for space-constrained areas and where the data rate is remarkably high.

In conclusion, AI on edge plays a vital role in enhancing the memory design for developing robust embedded systems. In the sphere of IoT, a strong data communication equipped with a reliable data storage and privacy mechanism should be there. For mobile robotics, a well-defined memory organisation determines the intelligence of a robotic system to prioritise a task. With these in place, the time to come is surely going to be really exciting as it will bring with itself new opportunities for the next generation of the advanced driver-assistance systems, artificial intelligence, and memoryless ecosystems. **EFY**

This article is based on the panel discussion ‘Memories: What’s New for Design Engineers?’ in November edition of India Technology Week 2020. The panel was chaired by Naresh Neelkantan, senior architect - system software at Sasken Technologies

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